IES | GENAI FOR ENGINEERS

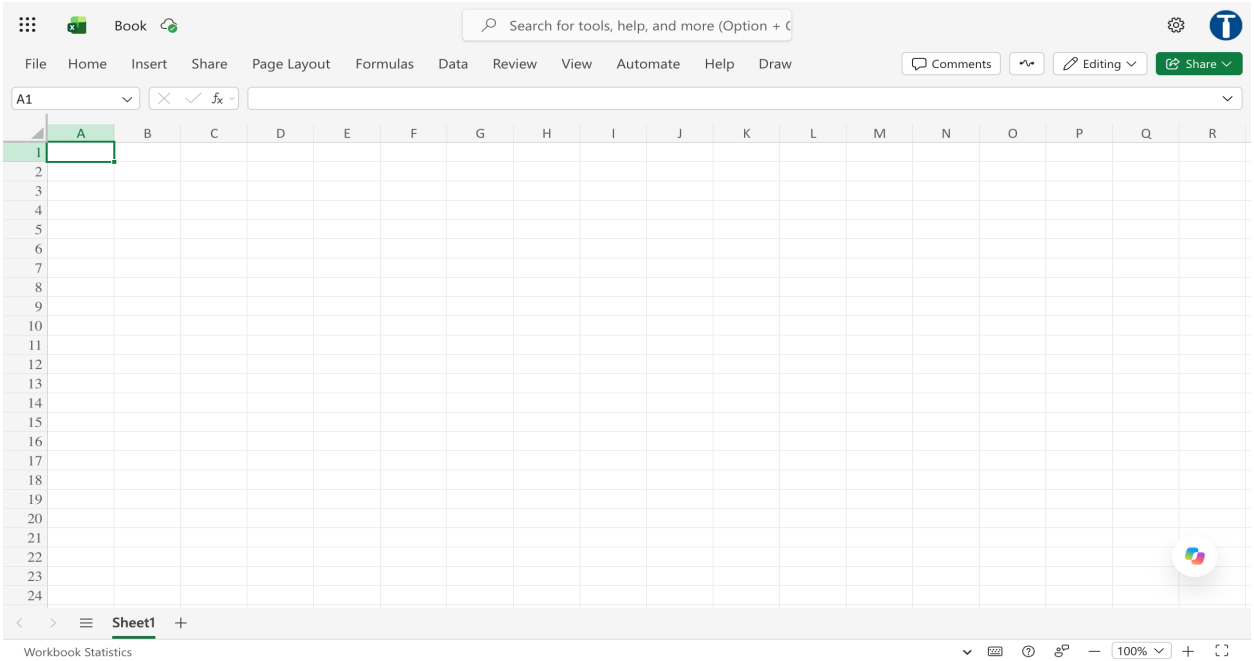
Lab 1 - Use Copilot in Excel to create an SPC review

25 minutes | Synthetic classroom case

A pump was stable during eight reference tests. Four later readings are monitored for unusual change. All values are synthetic training data.

1	Open Starter.xlsx and select a cell in PumpReadings.
2	Open Copilot in Excel.
3	Copy the prompt from the Copilot_Prompt sheet.
4	Ask Copilot to create the output and show its formulas.
5	Check one result yourself; correct the draft if needed.

Where to find Copilot



Build the prompt with RISE

Role - Input - Steps - Expectations

R - Role	Engineering data assistant
I - Input	Only PumpReadings; eight Baseline and four Monitoring rows
S - Steps	Calculate mean, standard deviation and limits; flag rows; create chart
E - Expectations	Show formulas; three sentences; no root-cause or safety claim

Paste the complete prompt into Copilot in Excel

Act as an engineering data assistant. Use only the named Excel table PumpReadings. Use the eight Baseline rows to calculate the mean, sample standard deviation, UCL = mean + 3 x standard deviation, and LCL = mean - 3 x standard deviation. Flag each Monitoring row as Below LCL, Above UCL, or Within limits. Create a line chart of all 12 readings with the centre line, UCL and LCL. Return a three-sentence interpretation. Do not claim a root cause or that the pump is safe. Show the formulas used.

Check one result yourself

Verify T10 = 59.4 L/h against the calculated LCL. A point below the LCL is a signal to investigate, not a diagnosis.

Expected Copilot draft

A baseline mean near 62.0 L/h, calculated limits, three low monitoring signals (T09, T10 and T12), and a short cautious interpretation.

Microsoft 365 Copilot links

Microsoft 365 Copilot

Get started with Copilot in Excel

How Copilot Chat works in Microsoft 365 apps